HEILBRONN UNIVERSITY
OF APPLIED SCIENCES

MASTER CERTIFICATE

Mr Kheireddine Azzez

who was born on 01/25/2000
in Teboulba
successfully graduated on 08/31/2024

from the degree program

Software Engineering

Master of Science (M.Sc.)

with a total grade of

very good (1.4)

The courses are indicated overleaf.

Heilbronn, 08/31/2024

President

Prof. Dr. Oliver Lenzen

Chairperson Examination Committee

Prof. Dr. Thomas Fankhauser

Mr Kheireddine Azzez Matr.-No. 215021

born on 01/25/2000 in Teboulba

Semester of studying: 4

Achieved credit points: 90

The examinations in individual subjects are graded as indicated below:

Advanced Study:

Title of the course	Grade	ECTS	App	Sem
Profile Advanced Software Engineering and Data Science	**1.0**	**24**		**st 2023**
Advanced Software Achitecture	1.0			wt 2022
Deep Learning	1.0			st 2023
DevOps and SecOps	1.0			st 2023
Information Visualisation	1.3			wt 2022
Intelligent Systems	1.0			wt 2022
Realtime Systems	1.0			wt 2022
Advanced Computer Science	**1.7**	**24**		**st 2023**
Cloud Computing	2.0			wt 2022
Usability Evaluation and Testing	2.0			wt 2022
Mixed Reality Lab	1.3			st 2023
Business Models für IT Innovation	2.0			wt 2022
Change and Innovation Management	1.3			st 2023
Management Methods/Leadership	1.7			wt 2022
Portfolio Optimization	**PA**	**12**		**wt 2022**
Scientific Writing	PA			wt 2022
German as a foreign language	1.0		AI	wt 2022
Advised Studying	PA			wt 2022
Master Thesis with Colloquium	**1.6**	**30**		**st 2024**
Master Colloquium	1.0			st 2024
Master Thesis *Enhancing User Experience and Productivity in Virtual Reality Programming Environments*	1.7			st 2024

Within the modules, individual examinations can be partially compensated according to the specifications of the study and examination regulations, which is why no ECTS points are presented for the individual grades. According to the ECTS User Guide, in such cases the total number of points required for a successful Master's degree is nevertheless assumed in accordance with the regular period of study.

(1) Description of the grading scale

The grading scale comprises 5 grades (with numerical equivalents):

Grade	
1,0-1,5	very good
1,6-2,5	good
2,6-3,5	satisfactory
3,6-4,0	sufficient
4,1-5,0	non sufficient

A minimum grade of "sufficient" (4.0) is required to pass.

(2) Status of examination

Examinations can achieve the following statuses with the meanings explained below:

PA	passed
F	failed
FF	finally failed

(3) Explanations of the course codes and the type of approval

The course code of an examination can be found in the study and examination regulations and in the corresponding module catalog.

Type of approval (App)	
AF	achievement provided at a foreign university
AI	achievement provided at other educational institution or other course of studies from Heilbronn University of Applied Sciences
AP	achievement provided at a partner university
C	compensation

(4) Explanation of the workload

1 full academic year	60 credits
1 Semester	30 credits
1 Trimester	20 credits